

Fundamental Trigonometric Identities



Using the Pythagorean identities

- 1 Given that $\sin \theta = \frac{3}{5}$ and θ lies in Quadrant II, find:
- a $\cos \theta$ b $\tan \theta$ c $\sec \theta$ d $\csc \theta$
- 2 Given that $\tan \theta = 2$ and θ lies in Quadrant III, find $\sin \theta$ and $\cos \theta$ in exact simplified form.
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Simplifying and verifying

- 3 Simplify each expression to a single trigonometric function:
- a $\frac{1 - \cos^2 x}{\sin x}$
- b $\sec x \cot x$
- c $\tan x \csc x$
- 4 Verify the identity $(1 + \tan^2 x) \cos^2 x = 1$.
- 5 Verify the identity $\frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$.
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Quotient and reciprocal identities

- 6 Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:
- a $\cot x \sin x$
- b $\frac{\sec x}{\csc x}$
- c $1 - (\sin^2 x \csc^2 x) \sin^2 x$
- 7 Simplify $\frac{\cos x}{1 - \sin x} - \tan x$ to a single term.
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More identity verifications

- 8 Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.
- 9 Verify the identity $\frac{1}{1 - \cos x} + \frac{1}{1 + \cos x} = 2 \csc^2 x$.
- 10 Show that $(\sin x + \cos x)^2 = 1 + 2 \sin x \cos x$.

- 11 Given that $\cos \theta = -\frac{8}{17}$ and θ lies in Quadrant III, find $\sin \theta$, $\tan \theta$, and $\cot \theta$ in exact simplified form.
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Identities with sums and products

- 12 Verify the identity $\sin x \cos x (\tan x + \cot x) = 1$.
- 13 Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.
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Identities involving all three Pythagorean forms

- 14 State all three forms of the Pythagorean identity (for sine/cosine, tangent/secant, and cotangent/cosecant).
- 15 Given $\cot \theta = \frac{5}{12}$ and θ in Quadrant I, find $\csc \theta$ and $\sin \theta$ using the cotangent-cosecant identity.
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Simplifying more complex expressions

- 16 Simplify $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$ using the sum-of-cubes factoring pattern.
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Verifying identities with a common denominator strategy

- 17 Verify $\tan x + \cot x = \sec x \csc x$.
- 18 Verify $\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x} = 1$.
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Numeric identity verification (checking a specific angle)

- 19 Verify the identity $1 + \tan^2 x = \sec^2 x$ numerically at $x = \frac{\pi}{6}$ by computing both sides.
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Identities with squared expressions

- 20 Simplify $(1 - \sin^2 x)(1 + \tan^2 x)$ to a constant.
- 21 Given that $\sec \theta = -\frac{13}{5}$ and θ lies in Quadrant II, use the Pythagorean identity $1 + \tan^2 \theta = \sec^2 \theta$ to find $\tan \theta$ exactly, and then find $\sin \theta$ using the quotient identity.

22 Verify the identity $\frac{\csc x - 1}{\cot x} = \frac{\cot x}{\csc x + 1}$ by cross-multiplying and using the Pythagorean identity.